DIGITAL FORENSICS INVESTIGATION REPORT

Memory Forensics — Linux CentOS 7 Compromise

Case Name	Seized Lab — Linux Memory Forensics Investigation
Case Number	2026-003
Examiner	Tej Chaute
Date	26/03/2026
Platform	CyberDefenders
Evidence File	dump.mem — SHA256: 3eb202361a8b6963330e8bd4c9d7e81ceff07d81e3c1bffb1b47ef678bf01717
Classification	Training

1. Case Summary

A memory forensics investigation was conducted on a Linux CentOS 7.7.1908 system (user: k3vin) to identify the cause of a security compromise. Analysis of the 0.99 GB memory dump (dump.mem) using Volatility 2 revealed that an attacker gained initial access via a reverse shell using ncat, established an interactive bash session, and maintained persistent access by injecting an SSH public key into /home/k3vin/.ssh/authorized_keys. A kernel-level rootkit (sysemptyrect) was also identified, confirming advanced attacker capabilities and stealthy persistence.

2. Evidence Examined

Item	Filename	Hash (SHA256)	File Size
001	dump.mem	3eb202361a8b6963330e8bd4c9d7e81ceff07d81e3c1bffb1b47ef678bf01717	0.99 GB

3. Tools Used

Tool	Version	Purpose
Volatility 2	2.6.1	Primary memory analysis — process listing, network, bash history, rootkit detection
strings	2.44	Extract raw ASCII/Unicode strings from memory dump
grep	3.11	Filter relevant strings and patterns from Volatility output
base64	9.5	Decode base64-encoded attacker payload found in bash history

4. Key Findings

Each finding is supported by at least one artifact. Source tool, Volatility plugin, and PID context are documented per item.

4.1 System Information

Field	Value
OS	CentOS Linux 7.7.1908 (Core)
Kernel	Linux 3.10.0-1062.el7.x86_64
User account	k3vin
Home directory	/home/k3vin
Evidence type	Full memory dump (.mem)
Timezone	UTC (memory dump — no local TZ confirmed)
Volatility profile	LinuxCentos7_3_10_1062x64

4.2 Artifact Table

#	Artifact Type	Location / Volatility Plugin	PID / Sequence	What It Proves
1	OS Version (string)	dump.mem → strings + grep 'CentOS'	N/A	System runs CentOS Linux 7.7.1908 — confirms correct Volatility profile needed
2	Suspicious process (ncat)	dump.mem → linux_pstree / linux_pslist	PID 2854	ncat running under k3vin — ncat is used for reverse shells;

#	Artifact Type	Location / Volatility Plugin	PID / Sequence	What It Proves
				presence confirms remote attacker session
3	Shell stabilization command	dump.mem → linux_psaux (PID 2887)	PID 2887 (bash)	Attacker executed python -c 'import pty; pty.spawn("/bin/bash")' — standard technique to upgrade a reverse shell to a fully interactive TTY
4	Reverse shell network connection	dump.mem → linux_netstat	PID 2854	Active TCP connection: local system → 192.168.49.1:12345 — confirms live C2 communication channel between victim and attacker machine
5	Bash history (encoded payload)	dump.mem → linux_bash (PID 2887)	Seq. 3 in bash history	Base64-encoded string found in bash history — decoded to attacker message confirming interactive command execution
6	Backdoor — git clone + pastebin payload	dump.mem → linux_bash (PID 2887)	Seq. 4–5 in bash history	Attacker cloned a repository and retrieved a base64-encoded payload from external Pastebin URL — confirms backdoor setup and remote payload delivery
7	SSH key injection (persistence)	dump.mem → linux_dump_map (PID 2887) → /home/k3vin/.ssh/authorized_keys	PID 2887	Attacker added RSA public key to authorized_keys — enables passwordless SSH login, rendering any password change ineffective
8	Rootkit — syscall table hook	dump.mem → linux_check_syscall	Kernel-level	Rootkit named sysemptyrect detected modifying the system call table — used for stealth and to hide

#	Artifact Type	Location / Volatility Plugin	PID / Sequence	What It Proves
				attacker processes/files from the OS
9	Rootkit encryption key (CRC65)	dump.mem → linux_lsmod -P	Kernel module	Encryption key '1337tibbartibbar' used by sysemptyrect rootkit — confirms advanced obfuscation; key needed to analyse rootkit behaviour further

4.3 Volatility Commands Run

All commands below were run against dump.mem with the confirmed profile. Linux Centos7_3_10_1062x64.

```
vol.py -f dump.mem --profile=LinuxCentos7_3_10_1062x64linux_pstree
vol.py -f dump.mem --profile=LinuxCentos7_3_10_1062x64 linux_psaux
vol.py -f dump.mem --profile=LinuxCentos7_3_10_1062x64 linux_netstat
vol.py -f dump.mem --profile=LinuxCentos7_3_10_1062x64 linux_bash
vol.py -f dump.mem --profile=LinuxCentos7_3_10_1062x64 linux_dump_map --pid=2887 -D ./output/
vol.py -f dump.mem --profile=LinuxCentos7_3_10_1062x64 linux_check_syscall
vol.py -f dump.mem --profile=LinuxCentos7_3_10_1062x64 linux_lsmod -P
strings dump.mem | grep -i 'centos\|linux' | head -20
base64 -d <<< '<encoded_string>'
```

5. Event Timeline

Reconstructed sequence of attacker actions based on artifact correlation. Memory forensics does not provide reliable wall-clock timestamps — sequence is inferred from process relationships, bash history order, and network state observed at time of dump.

Seq.	Phase	Artifact Source	Event Description
1	Initial Access	linux_netstat — PID 2854	Attacker established reverse shell to victim using ncat. Active TCP connection observed: victim → 192.168.49.1:12345
2	Execution	linux_psaux — PID 2887	Attacker upgraded shell to interactive TTY using python -c

Seq.	Phase	Artifact Source	Event Description
			'import pty; pty.spawn("/bin/bash")' to enable full command interaction
3	Discovery / Execution	linux_bash — PID 2887 (seq. 3)	Attacker ran base64-encoded command — decoded content confirms active interactive session and reconnaissance activity
4	Payload Delivery	linux_bash — PID 2887 (seq. 4–5)	Attacker cloned external git repository and retrieved base64-encoded payload from Pastebin — backdoor installation initiated
5	Persistence — SSH	linux_dump_map → authorized_keys	Attacker injected RSA public key into /home/k3vin/.ssh/authorized_keys — guaranteed re-entry even after password reset
6	Persistence — Rootkit	linux_check_syscall + linux_lsmod	Rootkit sysemptyrect loaded as kernel module — syscall table hooked for stealth. Encryption key '1337tibbartibbar' used for obfuscation
7	Post-Compromise	Full artifact correlation	System fully compromised: attacker has persistent SSH access, kernel-level rootkit hiding activity, and password change rendered ineffective

6. Conclusion

The investigation confirms the system (CentOS 7.7.1908, user k3vin) was fully compromised via a multi-stage attack:

Initial access (Artifact #2, #4): Attacker established a reverse shell using ncat (PID 2854) connecting to 192.168.49.1 on port 12345.

Execution (Artifact #3): Shell upgraded to interactive TTY via python pty.spawn, enabling full command-line control.

Payload delivery (Artifact #6): External payload retrieved from Pastebin via git clone, confirming planned and staged attack.

Persistence — SSH (Artifact #7): RSA public key injected into /home/k3vin/.ssh/authorized_keys, ensuring continued passwordless access regardless of password changes.

Persistence — Rootkit (Artifacts #8, #9): Kernel-level rootkit sysemptyrect loaded with encryption key '1337tibbartibbar', modifying the syscall table to hide attacker activity from the OS.
The combination of SSH key injection and a kernel rootkit indicates an experienced attacker with a clear objective to maintain long-term, stealthy access to the system.

7. Examiner Notes

During the investigation, initial challenges were encountered with Volatility profile setup including profile mismatch and missing dependencies. These were resolved by correctly identifying the kernel version from memory strings and selecting the matching Linux profile. This case provided strong practical insight into:

- SSH key-based persistence — a stealthy and very common real-world technique that survives password resets
- Kernel rootkit mechanics — syscall table hooking is advanced; detecting it with linux_check_syscall was a key learning point
- Memory-only evidence — no disk image was available, reinforcing that some attacker activity only exists in RAM and is lost on reboot
- Base64 encoding as obfuscation — attacker used encoding to evade basic string detection in logs